MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8587

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Beam power tetrode, pulse amplifier
See figure 1
Mounting position: Tube axis vertical, either end up
Weight: 38 pounds (17.24 kg) nominal

ABSOLUTE RATINGS: F = 600 MHz

Parameter:	Ef	eb	ec2	Ec1	ib	Ib	ic1	Ic1	ic2
Unit:	Vac	kv	kv	Vdc	a	mAdc	a	mAdc	a
Maximum:									
Anode pulsed amp class B (note 4), tp=15 µs:	1.00 (dc) 1.00 (60 Hz) 1.05 (400 Hz) (Note 1)	55 (Note 2)	2.2 (Note 3)	-400	80	320	15 (Note 5)	60	15
Anode pulsed amp class B (note 4), tp=70 µs:	1.00 (dc) 1.00 (60 Hz) 1.05 (400 Hz) (Note 1)	30 (Note 2)	2.2 (Note 3)	-400	30	500	3 (Note 5)	60	3
TEST CONDITIONS:	0.95 (Note 25)	---	---	---	---	---	---	---	---

ABSOLUTE RATINGS: F = 600 MHz

Parameter:	Ic2	Pi	Pp	tk	Cooling	Environmental requirements	Operating frequency	Fault protection	Du
Unit:	mAdc	kW	kW	sec	---	---	MHz	---	---
Maximum:									
Anode pulsed amp class B (note 4), tp=15 µs:	60	16	8	Note 6	Note 7	Note 9	174-600	Note 10	0.005
Anode pulsed amp class B (note 4), tp=70 µs:	60	9	5	Note 6	Note 7	Note 9	174-600	Note 10	0.023
TEST CONDITIONS:	---	---	---	Note 6	Note 8	---	---	Note 10	---

GENERAL:

Qualification - Required (see note 31)

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
---	Life test (1)	1,11, 12,13, 17	tp = 10 ±1 µs; po = 750 kw (min); F = 225 ±15 MHz	---	---	t	See note 12		---
---	Life-test (1) end point	11,17	tp = 10 ±1 µs; F = 225 ±15 MHz	---	---	po	750	---	kw
---	Life test (2)	1,12 13,17 27	tp = 60 ±1 µs; po = 200 kw (min); F = 425 ±15 MHz	---	---	t	See note 12		---
---	Life-test (2) end point	17,27	tp = 60 ±1 µs; F = 425 ±15 MHz	---	---	po	200	---	kw
---	Life test (3)	1,12, 13,17, 27	tp = 13 ±1 µs; po = 1,800 kw (min); F = 425 ±15 MHz	---	---	t	See note 12		---
---	Life-test (3) end point	17,27	tp = 13 ±1 µs; F = 425 ±15 MHz	---	---	po	1.8	---	Mw
---	Vibration	23,28		---	---	---	---	---	---
---	Shock	23,29		---	---	---	---	---	---
---	Vibration and shock test end point:								
---	Life test (1)	11		---	---	---	---	---	---
	Quality conformance inspection, part 1								
(B) 1155	Anode water pressure differential	14	No voltages applied	0.65	II	APD	---	25.0	psi
(B) 1155	Grid 1 water pressure differential	14	No voltages applied	0.65	II	GPD	---	6.0	psi
(B) 1155	Grid 2 water pressure differential	14	No voltages applied	0.65	II	SPD	---	8.0	psi
(B) 1155	Filament water pressure differential	14	No voltages applied	0.65	II	FPD	---	8.0	psi
(B) 1155	Cathode water pressure differential	14	No voltages applied	0.65	II	CPD	---	8.0	psi
1301	Filament current	-		0.65	II	If	460	530	Aac
(B) ---	Emission check	26	Peak power output; Ef/If = 450 A, 60 Hz; ec2 = 2.2 kv (max);Ib = 70a (max);F = 550 ±25 MHz	0.65	II	po	1.5	---	Mw
(B) 1266	Total grid current	20,24	Eb = 6.5 kVdc; Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; t = 5 minutes	0.65	II	Ic	---	-50	mAdc
---	Peak power output	26	eb = 50 kv (max); ec2 = 2.2 kv (max); F = 550 ±25 MHz; Ib = 70 a (max); t = 30 minutes	0.65	II	po	1.5	---	Mw
(B) ---	Power gain	19	Peak power output; rf drive/ po = 1.5 Mw	0.65	II	PG	150	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued								
1256	Electrode current (screen)	-	Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; Ebb = 6.5 kVdc	0.65	II	Ic2	-25	+25	mAdc
---	Input resonant frequency	21		0.65	II	Fi	222	250	MHz
---	Output resonant frequency	21		0.65	II	Fo	230	250	MHz
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	18							
---	Grid 2 to filament cathode	22		---	---	Cg2k	---	0.018	uF
	Anode to grid(1)	22	Use special shield adapter (RCA No. M-549-LT, or equivalent)	---	---	Cg1p	---	0.15	pF
1261	Electrode voltage (grid 1)	15	eb = 30 kv; ec2 = 2.0 kv; Ec1/Ib = 2.7a; F = 550±25 MHz	---	---	Ec1	---	-320	Vdc
	Quality conformance inspection, part 3								
---	Service-life guarantee	11,13	F = 174 to 550 MHz; t = 1,000 hours	---	---	---	---	---	---
---	Service-life guarantee end point:								
---	Peak power output	26,30	F = 550 ±25 MHz	---	---	po	1.2	---	Mw

NOTES:

1. Because the filament voltage, when operated near the maximum value, provides emission in excess of any requirements within tube ratings, during life the filament voltage shall be reduced to a value that will give adequate, but not excessive emission. Careful attention to maintaining the value consistent with adequate emission will result in conserving the life of the tube. The filament voltage shall be measured at the respective liquid coolant connections on the tube side of the threads. This procedure is essential in order to accurately measure the filament voltage. At 400 Hz some heating of the filament leads and rf cathode terminal (cathode header) occurs; this condition is not detrimental to tube operation or tube life.

2. Peak positive anode voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portion of the pulse. The magnitude of any spike on the anode voltage pulse shall not exceed the peak pulse value by more than 4,000 volts and the duration of any spike (measured at a point 70 percent above the peak pulse value) shall not exceed 2 µs. In the absence of rf output voltage, the peak pulse supply voltage may be exceeded by 20 percent by for not more than 10 consecutive pulses. A pressurized output cavity is required for operation at the maximum voltage specified herein to prevent external flashover at the insulating bushing. (See note 9.)

3. Peak positive screen voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portions of the pulse. The magnitude of any spike on the grid (2) voltage pulse shall not exceed the peak pulse value by more than 250 volts and the duration of any spike (measured at a point 70 percent above the peak pulse value) shall not exceed 2 µs.

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NOTES: -Continued

4. On time (tp) is defined as the sum of the durations of all the individual pulses which occur during any 3,000 µs interval.

5. Peak rectified grid(1) current.

6. After the application of filament voltage, a minimum of 90 seconds heating time is required before other electrode voltages can be applied.

7. Cooling requirements.

(a) Air cooling. Forced-air cooling of the ceramic bushing at the grid(1)seal and at the anode seal may be required in order to limit the temperature of the ceramic bushing at either seal to the specified maximum value of 150°C. Under such conditions, provisions shall be made for blowing air at the ceramic bushing through suitable openings in the coaxial-cylinder cavity circuit.

(b) Liquid cooling. Liquid cooling of the filament block, dc cathode block, grid(1)block, grid(2)block, and anode shall be required. When tube operation under low ambient temperatures is required, the recommended coolant is inert liquid FC75 (made by the Fluorochemical Division, Minnesota Mining and Manufacturing Co., or equivalent) but ethylene glycol mixed with water in the proportion of 60 percent ethylene glycol to 40 percent water by weight can be used. When the environmental temperature is above 0°C the coolant may be water. The liquid flow shall start before application of any voltages and preferably shall continue for several seconds after removal of all voltages. Interlocking of the liquid flow through each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate liquid flow. Liquid flow shall be as shown in table I.

TABLE I. Liquid flow.

	Absolute minimum flow (gpm)	Typical flow (gpm)	Maximum pressure differential for typical flow 1/ (psi)
With water:			
Through filament block	0.5	0.8	8
Through dc cathode block	0.5	0.8	8
Through grid 1 block	0.5	0.8	6
Through grid 2 block	0.5	0.8	8
Through anode in direction shown on figure 1:			
For anode dissipation up to 6 kw (average)	6	8	20
With inert liquid FC75, or equivalent:			
(with outlet liquid temperature of 30°C)			
Through filament block	1.0	1.2	20
Through dc cathode block	1.0	1.2	20
Through grid 1 block	1.0	1.2	14
Through grid 2 block	1.0	1.2	20
Through anode in direction shown on figure 1:			
For anode dissipation up to 6 kw (average)	8	10	25
With ethylene-glycol-water solution:			
Through filament block	1.0	1.2	18
Through dc cathode block	1.0	1.2	18
Through grid 1 block	1.0	1.2	12
Through grid 2 block	1.0	1.2	18
Through anode in direction shown on figure 1:			
For anode dissipation up to 6 kw (average)	8	10	30

1/ Directly across cooled element.

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NOTES: -Continued

8. The minimum waterflow to the anode cooling connection shall be 5 gpm. The minimum waterflow to the filament, filament-cathode, grid(1)and grid(2)cooling connections shall be 0.5 gpm. Waterflow shall start before the application of any voltages, and shall continue for at least 6 seconds after removal of all voltages. The outlet water temperature shall not exceed 70°C.

   Emergency conditions: Simultaneous interruption of cooling pump and tube electrode power is permissible.

9. Environmental requirements.

External gas pressure 1/	60 2/ psia maximum
Gage pressure at any inlet	100 psi maximum
Metal-surface temperature	100°C maximum
Ceramic-bushing temperature	150°C maximum
Outlet liquid temperature (any outlet) for water and FC75 (or equivalent)	70°C maximum
Outlet screen temperature (any outlet) for ethylene-glycol	60°C maximum
Storage temperature 3/	-65°C minimum

1/ Absolute pressure is the sum of the atmospheric pressure and gage pressure.
2/ This pressure is related to the output cavity pressurization as required to prevent corona or external flashover.
3/ The tube coolant ducts shall be free of liquid before storage or shipment of the tube to prevent damage from freezing, except that storage at -65°C is permissible with FC75 coolant or equivalent installed and at -45°C with ethylene-glycol solution installed.

10. High-speed electronic protective devices shall be used to prevent energy from the anode power supply, grid(1)power supply, and from the grid(2)power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the various pulse sources supplying the tube during the inter-pulse interval following the occurrence of a fault in any single pulse and shall be capable of meeting the requirements of the aluminum foil simulated fault tests. Thus, all pulses subsequent to a "fault" pulse will not be applied until the device is reset. (A fault is a condition in which the peak pulse currents rise 20 percent or more above the normal peak current level.) Additional protection against damage from an anode to grid(2)fault shall be provided to prevent the grid(2)voltage from rising to a value in excess of 3,000 volts during the fault pulse. Minimum reset time shall be 1 second.

(B)11. The tube shall be operated within the frequency range specified herein and the minimum power output specified herein shall be obtained when the operating conditions as specified and the following additional operating conditions are also applied: eb = 30.0 kv; ec2 = 2.0 kv; Ec1 = note 16; RF/Ib = 165 mAdc; Ic1 = -14 to +28 mAdc (max); Ic2 = 45 mAdc (max); t = 60 minutes.

12. Initial compliance with the statistical life test shown in table II shall be required. (X_0 = 1,000 hours, X_1 = 500 hours, a = 0.10, B = 0.10.)

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NOTES: -Continued

12. -Continued

TABLE II. Acceptance and rejection criteria.

Number of failures (r)	Total number of hours of burning	
	Reject	Accept
0	---	2,197
1	---	2,890
2	---	3,583
3	---	4,276
4	575	4,969
5	1,268	5,662
6	1,961	6,355
7	2,564	7,048
8	3,347	7,741
9	4,040	8,434
10	4,733	9,127
11	5,426	9,820
12	6,119	10,513

X_0 = Acceptable length of life in hours.

X_1 = Unacceptable length of life in hours.

a = Maximum risk of rejecting tubes if the length of life is X_0 or better.

B = Maximum risk of accepting the tubes if the length of life is X_1 or worse.

13. Water system requirements. In order to minimize maintenance, it is recommended that the water-cooling system be of the closed type utilizing water of acceptable quality as specified herein to prevent excessive deposition, corrosion, and electrolysis. Deposition of material not only restricts waterflow, but also prevents proper transfer of heat from the tube elements to the cooling water. Corrosion and electrolysis may destroy the tube elements, ducts, and fittings. Cooling water of acceptable quality shall have a specific resistivity in excess of 1 megohm-cm.

14. The waterflow of the anode cooling connection shall be 10 gpm. The waterflow to the filament, filament cathode, grid(1),and grid(2)cooling connections shall be 0.8 gpm.

15. The anode and grid(2) pulse length (tp) shall be 22 ±1 µs with a prr of 180 ±3.

16. The grid(1) voltage (Ec1) shall be adjusted for class B operation. Class B operation is defined as a negative E_{c1} voltage which is between 1/5 and 1/8 of the ec2 voltage.

17. Circuit adjustments are permissible. The rectified rf output pulse length (tp) shall be as specified with a prr of 300 ±5. Peak output power is calculated by dividing the average output power by the Du. The average output power shall be measured by a calorimetric method using a dummy water load, RCA Model MI No. 19198-A, or equivalent.

18. Any tube having one or more interelectrode capacitances greater than 100 pF shall be measured for all of its capacitances on suitable capacitance or impedence bridges operating at 1,000 Hz.

19. Power gain is the quotient of the peak output power and the peak rf input power. Peak rf input power shall be calculated by dividing the average rf input power by the Du. Average rf input power shall be measured with a calibrated directional coupler. The average rf input power is the difference between the forward and reverse values as indicated by the directional coupler. The grid circuit shall be adjusted for lowest VSWR in the drive line. Specification readings cannot be taken unless reflected power is less than 10 percent of the incident power. Calibration of the directional coupler shall be accomplished by the calorimetric method, or equivalent.

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NOTES: -Continued

20. With conditions as specified herein, grid(1)current (I_1) shall be recorded after specified time. Then, all voltages shall be removed except filament voltage and grid(1)voltage, and grid(1) leakage current (I_2) shall be recorded. Calculate Ic from the formula: $Ic = I_1 - I_2$.

21. The frequency of the sweep generator shall be adjusted to obtain resonance curve on oscilloscope as shown on figure 2. (Be sure tube is making good contact with cavity.) The marker oscillator frequency shall be varied until marker pip is located on peak of resonance curve. The resonant frequency as read on frequency meter shall be within limits specified herein. (The input cavity (R1) and output cavity (R2) shall be as shown on figures 3 and 4.)

22. Each element not under test shall be tied to ground. Two tubes per month shall be tested. Failure of one or more tubes will require 100 percent testing. Two tubes per month may be resumed after testing of 20 consecutive tubes without failure.

23. Test shall be performed with no voltages applied except as necessary to indicate shorts.

24. This test is to be the first test performed at the conclusion of the holding period.

25. Filament voltage is 60 Hz ac. After the application of filament voltage, a minimum of 90 seconds heating time is required before other electrode voltages can be applied.

26. The rectified rf output pulse length shall be 22 ±1 µs with a prr of 180 ±3. Peak output power is calculated by dividing the average output power by the Du. The average power shall be measured by a calorimetric method using an RCA dummy water load, Model MI No. 1918-A, or equivalent.

27. The tube shall be operated within the frequency range specified herein and the minimum power output specified shall be obtained when operating conditions are within maximum ratings for anode pulsed amplifier, class B at the tp as specified herein.

28. Test conditions.

(a) Resonance search.

Frequency: 5 to 55 Hz.
Duration of sweep: 5 minutes.
Number of sweeps: At least two (for example, 5 to 55 Hz and 55 to 5 Hz).
Displacement: 0.060 inch peak to peak.
Axes: X, Y, and Z.
Voltages: None except those of short checker.
Monitoring: Short checker as shown on figure 1041-1 of MIL-STD-1311. Velocity type pickup mounted on table to indicate table displacement. Crystal accelerometer mounted on tube to indicate tube displacement.

(b) Vibration fatigue.

Frequency: 50 Hz.
Duration: 12 hours per axis.
Displacement: 0.060 inch peak to peak.
Axes: X, Y, and Z.
Voltages: None except those of short checker.
Monitoring: Short check same as used in resonance search. Velocity type pickup and meter for table displacement.
Vibration equipment: Same as for resonance search.

29. The tube shall be tested in accordance with method 205 of MIL-STD-202. Use test condition A for height of drop.

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NOTES: -Continued

30. The tube shall be operated within the frequency range specified herein and the minimum power output specified herein shall be obtained when operating conditions are within conditions of peak power output test.

31. When qualification tests have been conducted for tube type 6952 of MIL-E-1/1106 and qualification approval has been granted, only the following tests need be conducted to obtain qualification approval for tube type 8587: Peak power output and electrode voltage (grid 1).

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N037)

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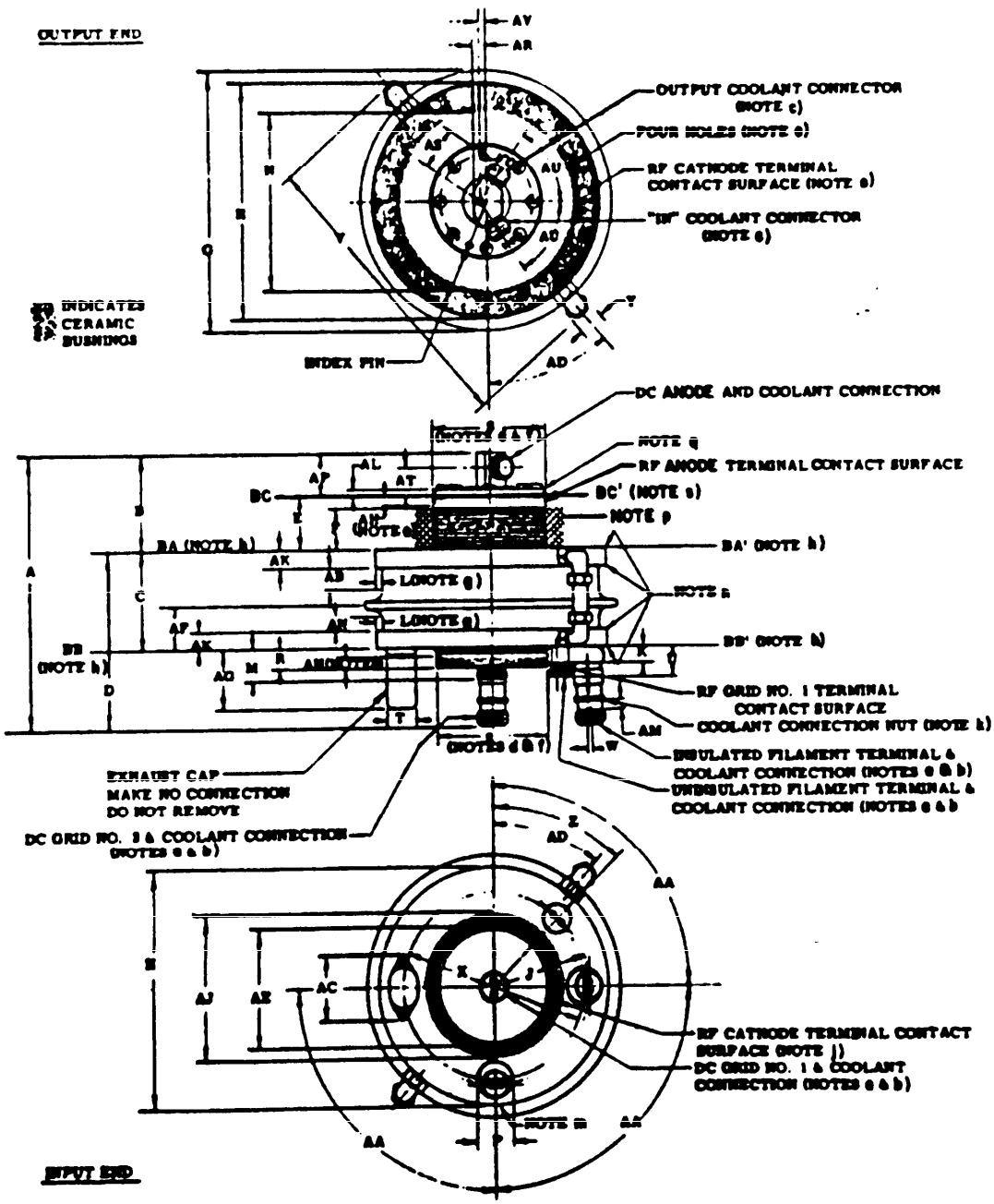


FIGURE 1. Outline drawing of electron tube type 8587.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			
E	1.86 (47.24)	1.89 (48.01)	
R	0.59 (14.99)	0.69 (17.53)	
S	3.74 (95.00)	3.76 (95.50)	d, f
Quality conformance inspection, part 2			
P	1.22 (30.99)	1.28 (32.51)	
V		11.25 (285.75)	
X	6.19 (157.23)	6.31 (160.27)	r
AB	0.50 (12.70)		g
AC	2.09 (53.09)	2.15 (54.61)	
AE		4.06 (103.12)	j
AF	1.38 (35.05)	1.50 (38.11)	
AH	0.38 (9.65)		e
AJ	4.95 (125.73)		j
AN	0.35 (8.89)		g
AR		0.46 (11.68)	
AV	0.19 (4.83)	0.27 (6.86)	
Ⓑ Quality conformance inspection, part 3			t
A	8.88 (225.55)	9.50 (241.30)	
B	3.20 (81.28)	3.32 (84.33)	
C	3.27 (83.06)	3.45 (87.63)	
D	5.68 (144.27)	6.18 (156.97)	
F	1.17 (29.72)		
G		8.75 (222.25)	
H	7.84 (199.13)	7.88 (200.15)	
K		0.59 (14.99)	
L	0.18 (4.57)		g
T	0.97 (24.64)	1.12 (25.45)	
U	0.83 (21.08)	0.95 (24.13)	
Y		0.68 (17.27)	
AG		2.00 (50.80)	
AK	0.56 (14.22)	0.63 (16.00)	
AL	0.243 (6.17)	0.257 (6.53)	
AM	0.09 (2.29)		
AP		1.42 (36.07)	
AS	1.34 (34.04)	1.42 (36.07)	
AT	0.92 (23.37)	0.98 (24.89)	
Reference dimensions			
M	1.09 (27.69)		r
N	6.00 (152.40)		e
J	3.25 (82.55) RAD		r
W	3°		r
Z	45°		r
AA	90°		r
AD	41°		r
AU	56°		

FIGURE 1. Outline drawing of electron tube type 8587 - Continued.

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NOTES:

a. Terminal has 1 inch-16, Unified Thread Class 2A fit, 0.38-inch long, and 2 holes 0.258 - 0.270 inch diameter spaced 0.438 inch on centers (quality conformance inspection, part 1 test).
b. The holes in the filament, grid No. 1, and grid No. 2 coolant terminal connections will accept the pins of the plug and cylinder combination gage shown in gage G1 shown on figure 5 (quality conformance inspection, part 1 test).
c. Coolant connectors are 0.750 inch-16 Unified Thread Class 2A fit, and will mate with Resistoflex Fitting No. R39531CSS-8, or equivalent.
d. The axis of the rf anode terminal contact surface is coincident with the axis of the grid No. 1 terminal contact surface within 0.05 inch.
e. Pressure from circuit contacts shall be exerted only over maximum length, AH, of designated contact areas of the anode or grid No. 1 terminals.
f. This diameter dimension is held only over a length of AH.
g. Dimension L applies over length AB and AN as indicated.
h. The contact surfaces, BA-BA' and BB-BB' of the rf cathode terminals shall be parallel within 0.06 inch.
j. Contact of the input end rf cathode terminal shall not be made at a diameter smaller than 4.06 inches nor greater than 4.95 inches.
k. To prevent excessive stress on the ceramic seal a 0.9375 inch open end wrench shall be used to permit gripping the terminal when removing or tightening the coolant connectors.
l. Contact of the output-end rf cathode terminal shall not be made at a diameter smaller than 6 inches. The pressure exerted for this rf contact shall be limited to that necessary for good electrical contact. The mechanical force for the cavity support and pressure seal shall be made at a diameter not less than 6 inches. On the output-end rf cathode terminal, there are four equally spaced 0.188 inch diameter holes on a circle having diameter of 6.75 inches. These holes are for tube manufacturing purposes only. Attention is called to the existence of these holes so that equipment designers can avoid making a pressure seal or electrical contact at points which are coincident with these holes. Mechanical clamping devices for the output cavity shall be designed so as to exert their clamping force across the outer edge of the output header flange.
m. Serial number shall be located on this surface between dc grid No. 2 and filament-section No. 1 connectors.
n. Corners may be rounded or chamfered not to exceed 0.05 inch.
p. Keep this 5 inch diameter annular-volume region clear. Do not allow circuit components (conductors or insulators) to protrude into this region.
q. This edge shall be rounded to a 0.03125 inch radius. The diameter of this coolant seal plate shall be 3.710 inches maximum. The eight holes in this plate shall be equally spaced on a circle having a diameter of 3.14 - 3.16 inches.
r. When the right angle separator is replaced by separator used on tube type 6952 (see MIL-E-1/1106), the output-end shall accept the output end of gage G-3 shown on figure 6. Proper acceptance is obtained when gage is seated on the output end rf cathode terminal contact surface. The gage is properly seated when a 0.015 inch thickness gage 0.50 inch wide will not enter between the gage and the rf cathode contact surface. The pins on the 1.500 inch movable section of the output end of gage G-3 shall enter the anode coolant holes such that the height slot represented by the 2.098 inch dimension is below the shoulder of the gage and the 2.219 inch dimension is visible above the shoulder.

FIGURE 1. Outline drawing of electron tube type 8587 - Continued.

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(1) The input end shall accept the input end of gage G-3 shown on figure 6. Proper acceptance is obtained when the gage is seated on the input-end rf cathode terminal surface and the projecting guide pins of the output end engage the mating holes of the input end. The gage is properly seated when a 0.015 inch thickness gage 0.50 inch wide will not enter between the gage and rf cathode terminal contact surface. The pins on the 0.875 inch movable section of the input end shall enter the grid coolant holes such that the height slot represented by the 2.379 inch dimension is below the shoulder of the gage and the 2.559 inch dimension is visible above the shoulder.

(2) When gage G-3 is properly seated:

(a) Dimensions J, M, W, X, Z, AA, and AD shall vary only within the limits of the gage.
(b) The centerline through the dc grid No. 1 coolant connection holes will have a maximum angle of 2 degrees with respect to the centerline of the dc anode coolant connection holes.
(c) The axis of the rf anode terminal contact surface is coincident with the axis of the grid No. 1 terminal contact surface within 0.05 inch.

s. The centerline of the inlet and outlet coolant connectors shall be parallel to the plane of the surface BC-BC' within 0.012 inch.

t. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8587 - Continued.

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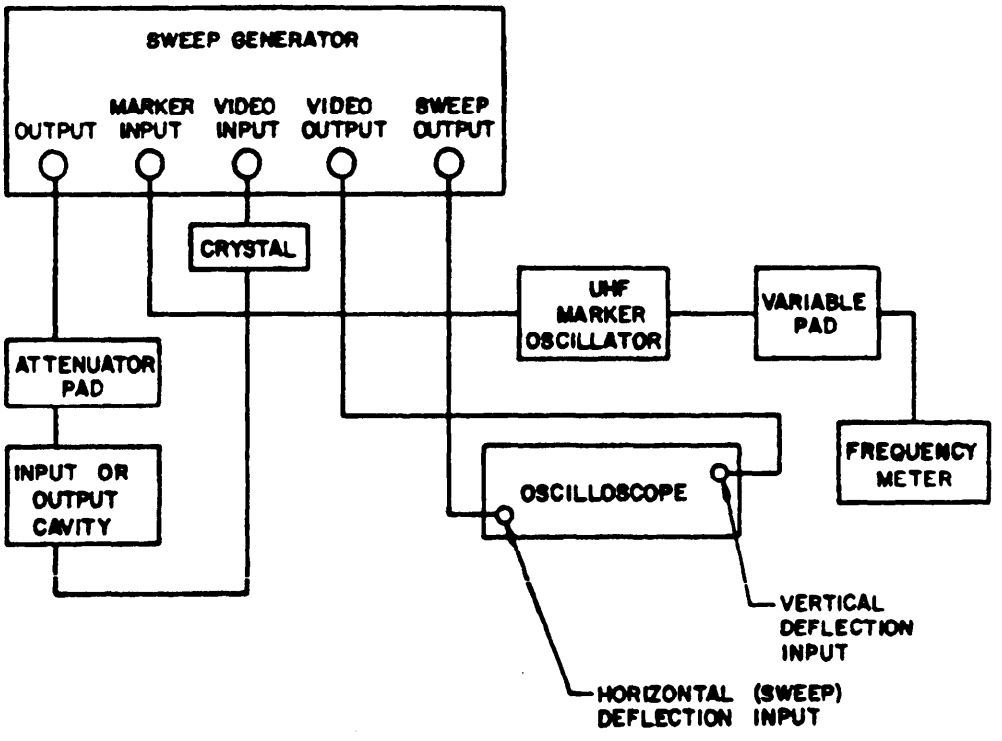
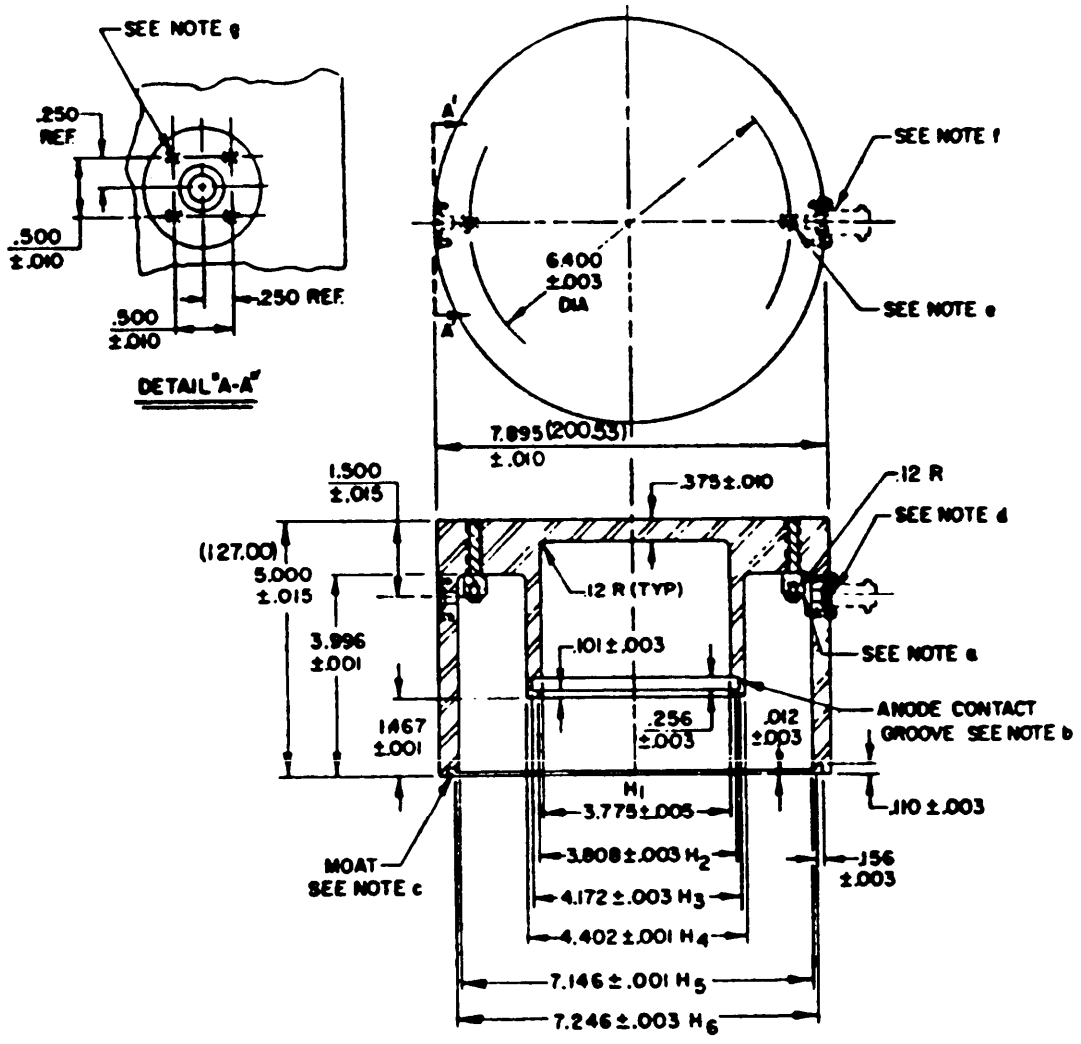


FIGURE 2. Typical block diagram of test circuit.

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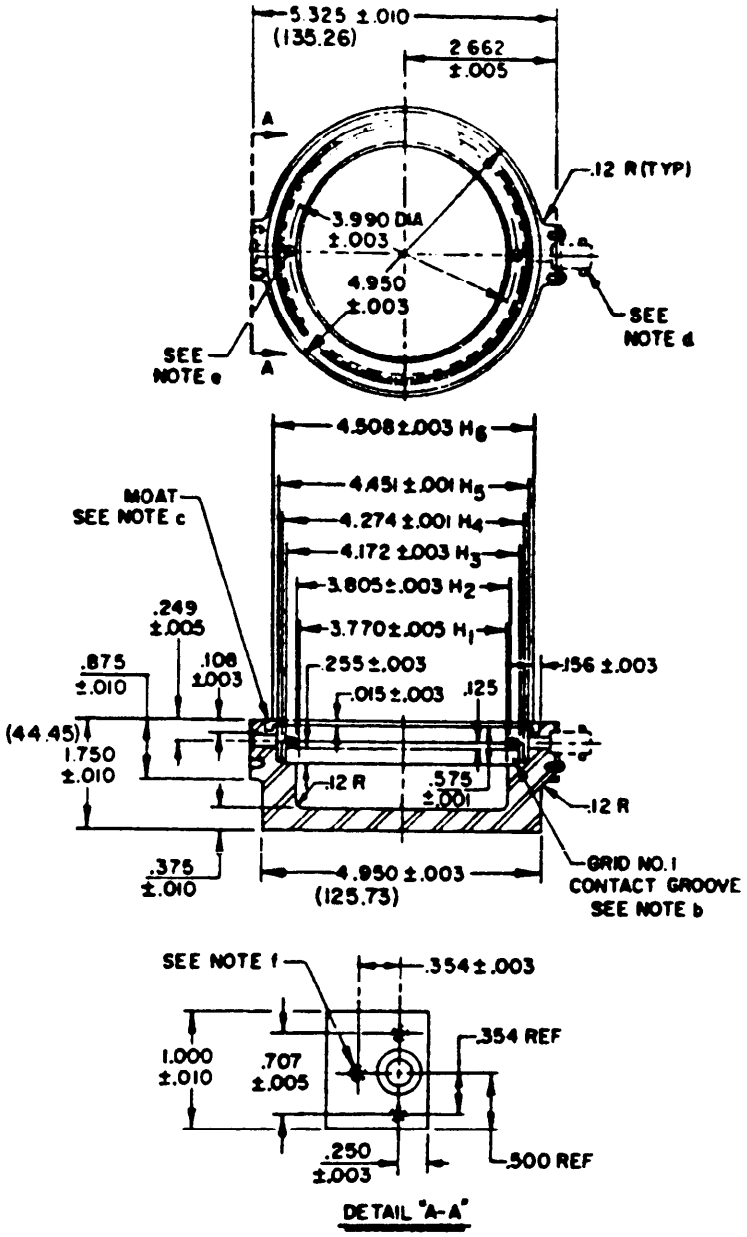


NOTES:

a. 2 studs. 1.5 inch long threaded .25 inch-20 NC by 1.0 inch with .375 inch diameter head. Head of stud shall be drilled and tapped .250 inch deep for .138-32 RH machine screw. Use standard .152 inch flat washer under screw.
b. Anode contact groove shall be filled with the following spring: .020 inch diameter wire wound to give a .250 inch O.D. at 35 TPI, 14.250 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the rf cathode terminal contact surface.
d. 2 holes. Drill .219 inch diameter (thru), C' bore .375 inch diameter by .125 inch deep. Spot face 1.0 inch diameter by .062 inch deep.
e. 2 holes. Drill and tap for .25 inch-20 NC screws.
f. Location of A-N connector -UG-290/U (two connectors 180 degrees apart).
g. 4 holes. Drill and tap for #3-48 screws by .25 inch deep.
h. Cylindrical holes H1 to H6 have axes coincident within .001 inch.

FIGURE 3. Output cavity required for resonant frequency test.

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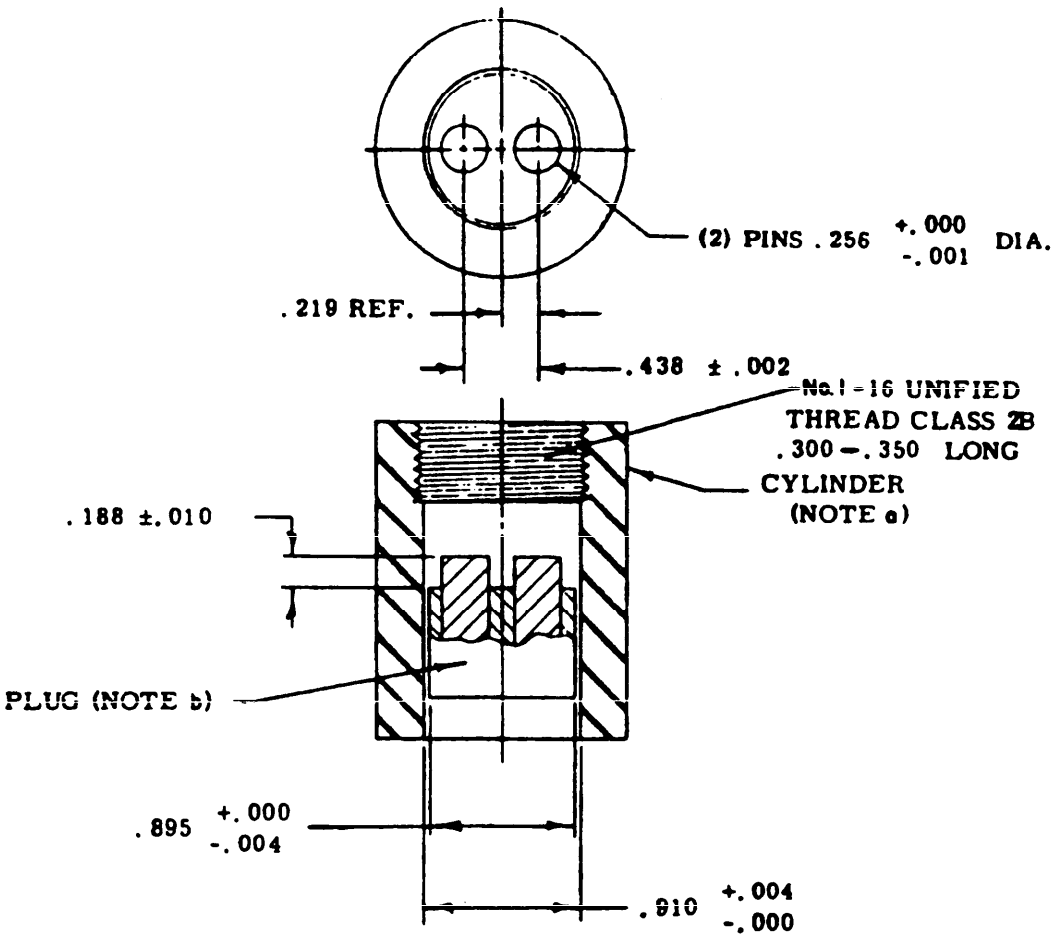


NOTES:

a. 2 holes, drill .219 inch diameter (thru) C' bore .375 inch diameter by .062 inch deep.
b. Grid No. 1 contact groove to be filled with the following spring: .020 inch diameter wire wound to give a .250 inch O.D. at 35 TPI, 14.250 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the RF cathode terminal contact surface.
d. Location of A-N connector -UG-290/U (two connectors 180 degrees apart).
e. 2 holes. Drill and tap for #3-48 screws.
f. 3 holes. Drill and tap for #3-48 screws by .187 inch deep.
g. Cylindrical holes H1 to H6 have axes coincident within .001 inch.

FIGURE 4. Input cavity required for resonant frequency test.

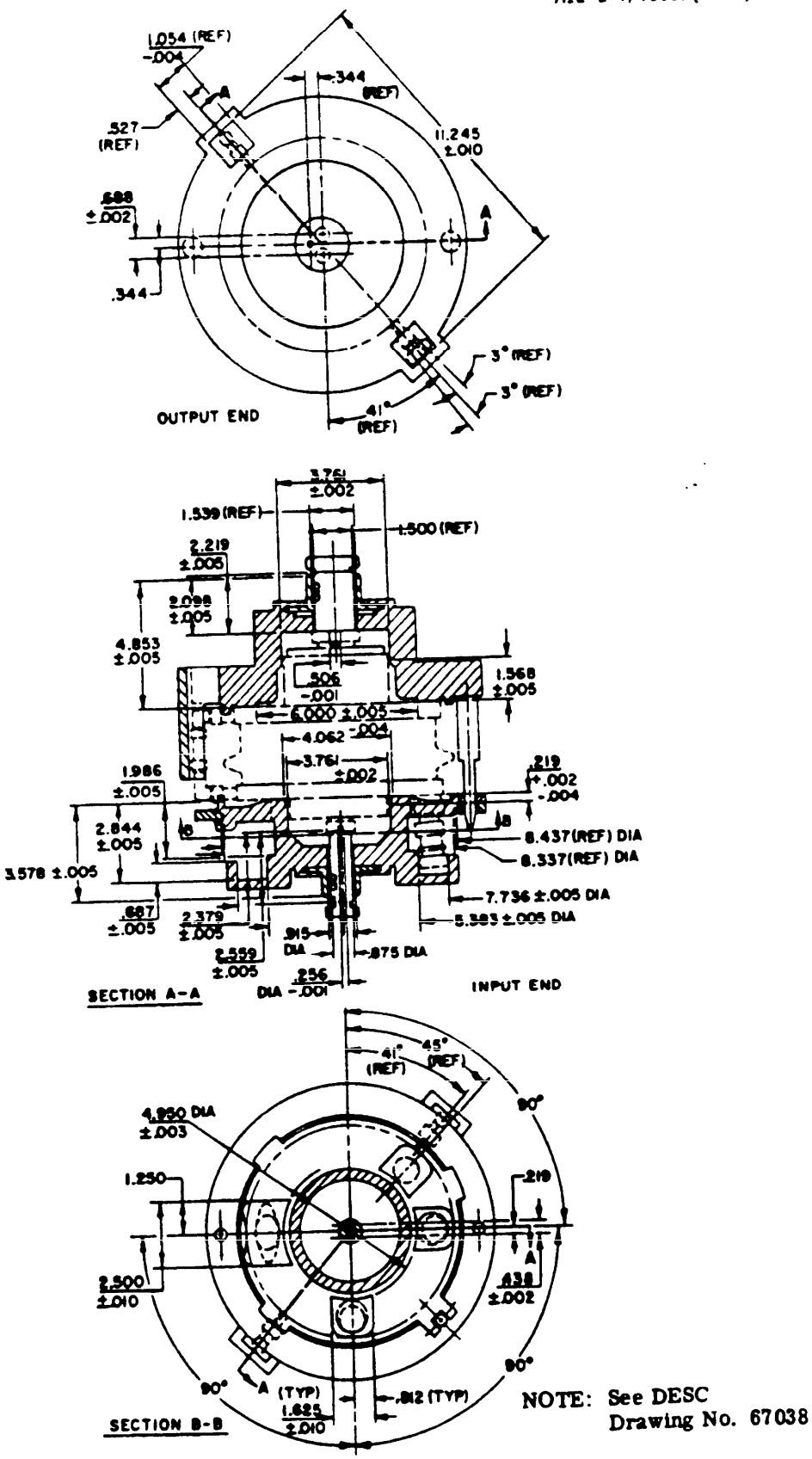
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NOTES:
a. Axis of the cylindrical holes and threaded section shall be coincident within 0.002 inch.
b. Plug sides and pin sides shall be parallel within 0.001 inch.

FIGURE 5. Gage G-1.

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NOTE: See DESC Drawing No. 67038

FIGURE 6. Gage G-3.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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